

Pre-Board Examination (2024-25)**Class- X****Time: 3 Hrs****Subject: Mathematics Basic****MM: 80****General Instructions:**

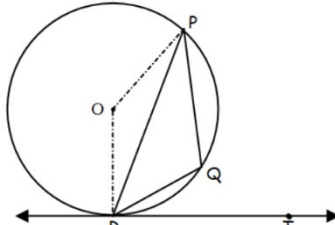
Read the following instructions carefully and follow them:

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take $\pi = 22/7$ wherever required if not stated.
11. Use of calculators is not allowed.

Section A

Section A consists of 20 questions of 1 mark each.

1.	If two positive integers p and q are written as $p = x^3y^2z$ and $q = xy^3z$; x, y and z are prime numbers, then LCM (p, q) is (A) xyz (B) xy^2 (C) x^3y^3z (D) xyz	1
2.	A quadratic polynomial, whose zeroes are -3 and 4, is (A) $x^2 - x + 12$ (B) $x^2 + x + 12$ (C) $x^2 - x - 12$ (D) $x^2 + x - 12$	1
3.	Values of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots is (A) 0 only (B) 4 (C) 8 only (D) 0, 8	1
4.	The list of numbers - 10, - 6, - 2, 2,... is (A) an AP with d = - 16 (B) an AP with d = 4 (C) an AP with d = - 4 (D) not an AP	1
5.	D and E are respectively the points on the sides AB and AC of a triangle ABC such that AD = 2 cm, BD = 3 cm, BC = 7.5 cm and DE // BC. Then, length of DE (in cm) is (A) 2.5 (B) 3 (C) 5 (D) 6	1
6.	The mid-point of the line segment joining the points A (-4, 8) and B (8, - 4) is (A) (2, 2) (B) (- 2, - 6) (C) (-2, -2) (D) (4, 2)	1
7.	The distance of the point P (2, 3) from the x-axis is (A) 2 (B) 3 (C) 1 (D) 5	1
8.	If one of the zeroes of the quadratic polynomial $(k-1)x^2 + kx + 1$ is -3, then the value of k is (A) $\frac{4}{3}$ (B) $\frac{-4}{3}$ (C) $\frac{2}{3}$ (D) $\frac{-2}{3}$	1
9.	A card is selected at random from a well shuffled deck of 52 playing cards. The probability of its being an ace is (A) $\frac{4}{13}$ (B) $\frac{3}{13}$ (C) $\frac{2}{13}$ (D) $\frac{1}{13}$	1
10.	If $\sin A = \frac{1}{2}$, then the value of cosec A is (A) 2 (B) $\sqrt{2}$ (C) $\frac{2}{\sqrt{3}}$ (D) Not defined	1
11.	A pole 6 m high casts a shadow 6 m long on the ground, then the Sun's elevation is (A) 60° (B) 45° (C) 30° (D) 90°	1
12.	In given Fig., PR is a chord of a circle and RT is the tangent at R such that $\angle PRT = 70^\circ$. Then	1

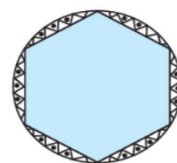
	$\angle PQR$ is equal to (A) 140° (B) 120° (C) 150° (D) 110°																
13.	If two tangents inclined at an angle 90° are drawn to a circle of radius 3 cm, then length of each tangent is equal to (A) $\frac{3\sqrt{3}}{2}$ cm (B) 6 cm (C) 3 cm (D) $3\sqrt{3}$ cm		1														
14.	If θ is the angle (in degrees) of a sector of a circle of radius r, then area of the sector is (A) $\frac{\pi r^2 \theta}{360}$ (B) $\frac{\pi r^2 \theta}{180}$ (C) $\frac{2\pi r \theta}{360}$ (D) $\frac{2\pi r \theta}{180}$		1														
15.	A cylindrical pencil sharpened at one edge is the combination of (A) a cone and a cylinder (B) a hemisphere and a cone (C) a hemisphere and a cylinder (D) two cylinders.		1														
16.	A right circular cylinder of radius r cm and height h cm ($h > 2r$) just encloses a sphere of diameter (A) r cm (B) 2r cm (C) h cm (D) 2h cm		1														
17.	Construction of a cumulative frequency table is useful in determining the (A) mean (B) median (C) mode Solution (D) all the above three measures		1														
18.	Consider the following frequency distribution of the heights of 60 students of a class: <table border="1" data-bbox="229 960 1362 1046"><tr><td>Height (in cm)</td><td>150-155</td><td>155-160</td><td>160-165</td><td>165-170</td><td>170-175</td><td>175-180</td></tr><tr><td>No. of students</td><td>15</td><td>13</td><td>10</td><td>8</td><td>9</td><td>5</td></tr></table> The sum of the lower limit of the modal class and upper limit of the median class is (A) 310 (B) 315 (C) 320 (D) 330	Height (in cm)	150-155	155-160	160-165	165-170	170-175	175-180	No. of students	15	13	10	8	9	5		1
Height (in cm)	150-155	155-160	160-165	165-170	170-175	175-180											
No. of students	15	13	10	8	9	5											
	DIRECTION: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) B) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) C) Assertion (A) is true but reason (R) is false. D) Assertion (A) is false but reason (R) is true.																
19.	Assertion (A): The pair of equations $x + 2y - \frac{1}{2} = 0$ and $-3x - 6y + 1 = 0$ have no solution Reason (R): Graphical representation of two parallel lines will have no point in common.		1														
20.	Assertion (A): A bag contains 3 red balls, 5 white balls and 7 black balls. The probability that a ball drawn from the bag at random neither red nor black will be $\frac{10}{15}$ i.e. $\frac{2}{3}$. Reason (R): The theoretical (or classical) probability of an event E [denoted by P(E)] is given by $P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Number of all possible outcomes of the experiment}}$		1														
Section B																	
Section B consists of 5 questions of 2 marks each.																	
21.	Can two numbers have 6 as their HCF and 100 as their LCM? Give reasons. OR If $\text{HCF}(119, 221) = 17$, then find their LCM.		2														
22.	Find the ratio in which the line segment joining the points $(-3, 10)$ and $(6, -8)$ is divided by $(-1, 6)$.		2														

23.	Find a relation between x and y such that the point P(x, y) is equidistant from the point A(3, 6) and B(-3, 4).	2
24.	If $\cos(A + B) = \frac{1}{2}$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$; $0^\circ < A + B \leq 90^\circ$; $A > B$, then find the values of A and B.	2
25.	A bag contains slips numbered from 10 to 90. If Fatima chooses a slip at random from the bag, find the probability of getting a number which is multiple of 5. (OR) A coin is tossed two times. Find the probability of getting at most one head.	2

Section C

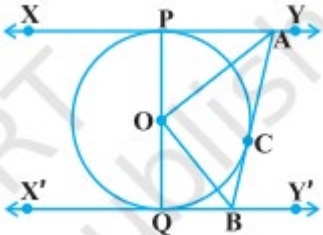
Section C consists of 6 questions of 3 marks each.

26.	Prove that $\sqrt{3}$ is irrational.	3
27.	Find the zeroes of the polynomial $6x^2 + 5\sqrt{3}x - 3$ by factorisation method and verify the relation between the zeroes and the coefficients of the polynomial.	3
28.	If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Prove.	3
29.	The altitude of a right triangle is 17 cm less than its base. If the hypotenuse is 25 cm, find the other two sides. OR Check whether the equation $6x^2 - 7x + 2 = 0$ has real roots, and if it has, find them by the method of quadratic formula.	3
30.	Prove that: $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$ OR Prove that: $(\sin \alpha + \cos \alpha)(\tan \alpha + \cot \alpha) = \sec \alpha + \operatorname{cosec} \alpha$	3
31.	A round table cover has six equal designs as shown in given Fig. If the radius of the cover is 42 cm, find the cost of making the designs at the rate of ₹20 per 100cm^2 . (Use $\sqrt{3} = 1.7$)	3



Section D

Section D consists of 4 questions of 5 marks each.

32.	Solve the pair of equations $3x - 2y - 6 = 0$ and $2x - 3y + 6 = 0$ graphically and find the area of triangle formed between the lines and the x-axis.	5																
33.	From a balloon vertically above a straight road, the angles of depression of two cars at an instant are found to be 45° and 60°. If the cars are 100 m apart, find the height of the balloon. OR The angle of elevation of the top of a tower from certain point is 30°. If the observer moves 20 metres towards the tower, the angle of elevation of the top increases by 15°. Find the height of the tower.	5																
34.	 In figure, XY and X'Y' are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X'Y' at B. Prove that $\angle AOB = 90^\circ$.	5																
35.	If <table border="1" data-bbox="245 2024 1321 2112"><tr><td>Class interval</td><td>0-10</td><td>10-20</td><td>20-30</td><td>30-40</td><td>40-50</td><td>50-60</td><td>Total</td></tr><tr><td>Frequency</td><td>5</td><td>x</td><td>20</td><td>15</td><td>y</td><td>5</td><td>60</td></tr></table> the	Class interval	0-10	10-20	20-30	30-40	40-50	50-60	Total	Frequency	5	x	20	15	y	5	60	5
Class interval	0-10	10-20	20-30	30-40	40-50	50-60	Total											
Frequency	5	x	20	15	y	5	60											

median of the distribution given below is 28.5, find the values of x and y.

(OR)

100 surnames were randomly picked up from a local telephone directory and the frequency distribution of the number of letters in English Alphabets in the surnames was obtained as follows:

Number of letters	1-4	4-7	7-10	10-13	13-16	16-19
Number of surnames	6	30	40	16	4	4

Determine the median number of letters in surnames. Find the mean number of the letters in the surnames? Also, find the modal size of the surnames.

Section E

Section E consists of 3 case study based questions of 4 marks each.

36. The students of a school decided to beautify the school on the Annual Day by fixing colourful flags on the straight passage of the school. They have 27 flags to be fixed at intervals of every 2 m. The flags are stored at the position of the middle most flag. Akshra was given the responsibility of placing the flags. Akshra kept her books where the flags were stored. She could carry only one flag at a time.



- How many flags are fixed each side of the centre flag? 1
- What is distance of 10th flag on either side from the centre? 1
- How much distance did she cover in completing this job and returning back to collect her books?

OR

What is the maximum distance from centre she travelled carrying a flag? 2

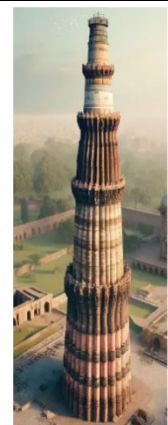
37. Tanisha was trying to find the height of QutubMinar near her house in Delhi. She is using the properties of similar triangles. The height of her mother is 1.6m. When her mother casts a shadow of 1.2m long on the ground, at the same time, QutubMinar casts a shadow of 54 m long on the ground as shown in the figure.

Based on the above information, answer the following questions.

- Which concept is used here to find various lengths? 1
- What is ratio of her mother's height to her shadow's length? 1
- What will be the length of shadow of Alina's house if it is 20m in height?

OR

What is height of Qutub Minar? 2



4

4

38. The Great Stupa at Sanchi is one of the oldest stone structures in India, and an important monument of Indian Architecture. It was originally commissioned by the emperor Ashoka in the 3rd century BCE. Its nucleus was a simple hemispherical brick structure built over the relics of the Buddha. It is a perfect example of combination of solid figures. A big hemispherical dome having base radius 14m with a cuboidal structure mounted on it of dimensions 8m x 6m x 4m. (Take $\pi = 22/7$)

- Write the formula to find the Volume of Sphere. 1
- Which shape the Paridakshinapatha is forming? 1
- Find the volume of the stupa.

OR

Find the cloth require to cover the hemispherical dome leaving cuboidal portion.

2

